

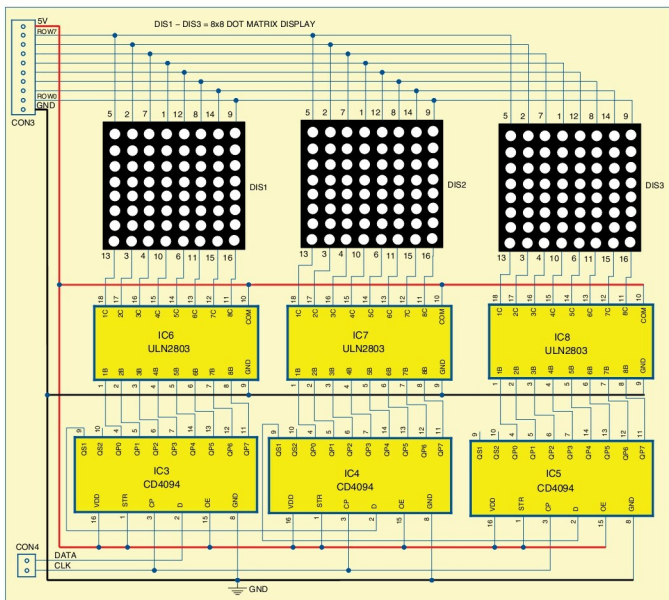


Fig. 4: Display unit

rows and five columns of the array are controlled through a microcontroller.

If we want to display alphabet A, we will first select column C1 (which means C1 is pulled low in this case) and deselect other columns by blocking their ground paths (one way of doing that is by pulling C2 through C5 pins to logic high). Now, the first column is active, and you need to turn on the LEDs in rows R2 through R7 of this column, which can be done by applying forward-bias voltages to these rows.

Next, select column C2 (and deselect all other columns) and apply

forward-bias voltages to resistors R1 and R5, and similarly for columns C3 and C4. Then, activate column C5 by pulling it down and deselect other columns, and apply forward-bias voltages to LEDs in rows R2 through R7.

By repeating these steps quickly (>100 times per second), and turning on the respective LEDs in each row of that column, the persistence of vision comes into play and we perceive the displayed image of the alphabet A as still.

You must have noticed that across each row one pin is sourcing the current for only one LED at a time, but

a column pin may have to sink the currents from more than one LED. For example, column C1 should be able to sink the current from six LEDs while displaying alphabet A. A microcontroller has low sourcing as well as sinking capabilities. To obviate this limitation, external transistor arrays or buffers are used. In this project, pnp transistor BC558 (T1-T8) has been used for this purpose. In the circuit an 8x8 dot-matrix has been used.

Brief description of the ICs used

IC1 is a 7805, 5V regulator IC, which